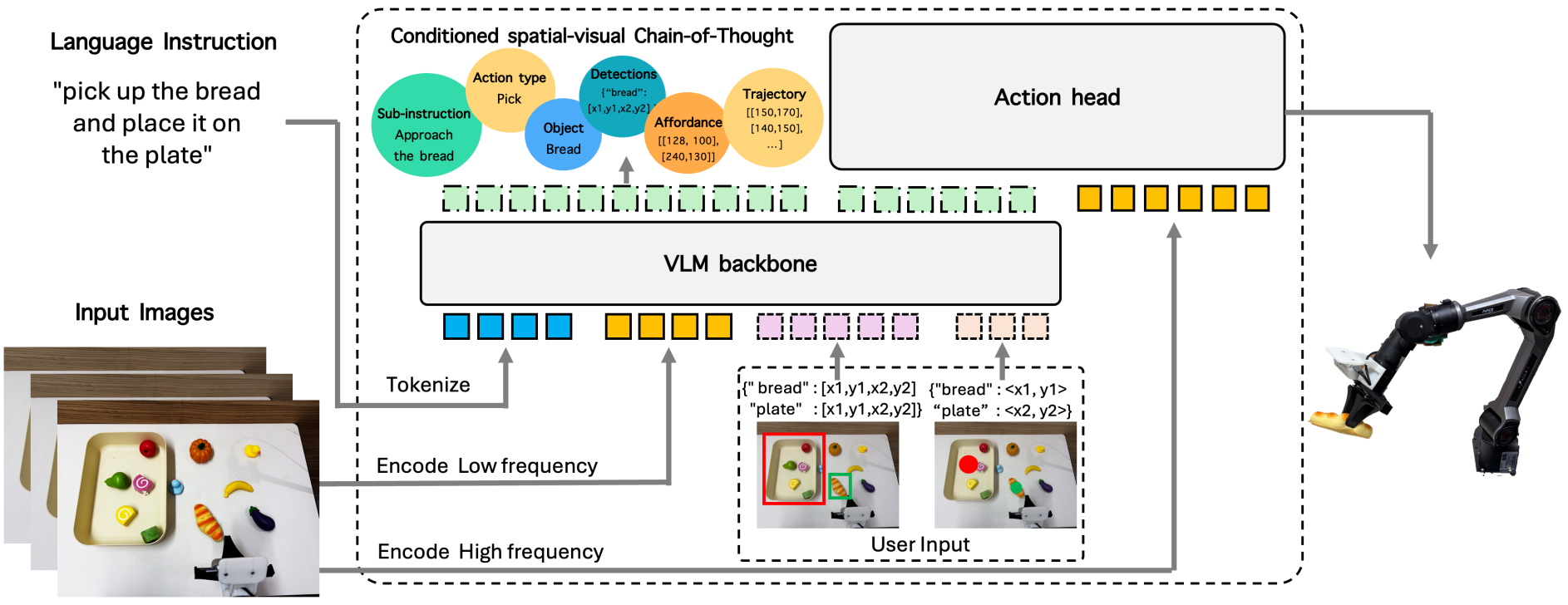
Language Instruction
"pick up the bread and place it on the plate"
Input Images
Conditioned spatial-visual Chain-of-Thought
Sub-instruction
Approach the bread
Action type
Pick
Object
Bread
Detections
{"bread": [x1,y1,x2,y2]}
Affordance
[[128, 100], [240,130]]
Trajectory
[[150,170], [140,150], ...]
Action head
VLM backbone
Tokenize
Encode Low frequency
Encode High frequency
{" bread" : [x1,y1,x2,y2]
"plate" : [x1,y1,x2,y2]}
{"bread" : <x1, y1>
"plate" : <x2, y2>}
User Input